A030 – Compact Geometry: T^4, Z_3-Orbifold, Winding Classes

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.1 Purpose

A020 stipulated the space T^4/Z_3. Here we develop what this stipulation contains – and, equally important, what it does *not* contain.

The distinction is practical. A large part of the objections to the theory target a picture it does not hold: that space is divided into cells, that there is a tiling, that discreteness comes from a partition. None of this is the case, and the reason is explained in this document.

.2 Compactness and the Discrete Spectrum

[B] The only consequence of compactness that matters is this: the Laplace operator on a compact manifold has a discrete spectrum. The eigenvalues form a countable, upward unbounded sequence; between two adjacent eigenvalues there is nothing.

This is elementary spectral theory and no claim of FFGFT. What FFGFT claims is the assignment: that physical masses are precisely these eigenvalues, and not numbers inserted into the model from outside.

[B] It follows immediately that there is a smallest permitted step. A state cannot lie a little between two modes. This discreteness is spectral, not spatial – it comes from the boundarylessness and finiteness of the space, not from a partition of it.

.3 Why a Torus and Not a Sphere

[B] The torus has a non-trivial fundamental group,

$$\pi_1(T^4) \cong \mathbb{Z}^4,$$

the sphere does not: $\pi_1(S^n) = 0$ for $n \geq 2$.

The difference is the entire reason for the choice. On the torus there are closed paths that cannot be contracted to a point, and every such path carries four integers – how many times it goes around in each of the four directions. These winding numbers are:

- **integer**: there is no half-winding;
- **coordinate-free**: they change under no reparametrisation;
- **topological**: hence robust against any continuous deformation.

[B] On a simply connected surface none of these quantities exists. Replacing the torus loses not a detail of the construction but the carrier of the entire information concept (A180) and the explanation of Bell correlations as topological connection (A160).

.4 What the Elementary Objects Are

[STIPULATION] The elementary discrete objects of the theory are not spatial regions but invariants: winding classes and eigenmodes.

This sentence answers an entire family of objections at once and therefore deserves an explicit counterpart.

Against a space-cell ontology the usual objections are correct: where does the tiling originate? Why this orientation? What happens under a coordinate change? How can an entropy depending on an arbitrary partition be physical?

[B] These objections run empty for a spectral-topological individuation. A winding class has no location – it is a property of a closed path, not of a region. An eigenmode has no origin – it is global. An entropy formed as a function of the spectrum is automatically invariant under origin, orientation, and coordinate choice, because the spectrum is.

[B] The torus is moreover boundaryless, $\partial T^4 = \emptyset$. This also excludes a surface-based definition of fundamental units: there is no distinguished surface. A surface in this geometry can only be a projection surface, and projection is lossy (A090).

.5 The Four Circles

[B] Four circles, not five: because $\tilde{T} \cdot m = 1$ binds time to mass, time needs no circle of its own. The minimum number is therefore four, and in this sense the four is derived, not chosen.

In the compact picture the four circles are *equivalent*. There is no space-time distinction as long as the space is compact – time is written just like the three other directions. The familiar asymmetry between space and time arises only when one circle is unrolled, and that is a measurement process, not a geometric one (A090).

[STIPULATION] That the experienced structure is $3 + 1$ remains empirical input (register P34). The geometry says that exactly one circle is unrolled; it does not say which one, and it does not say why it is one.

.6 The Z_3 Identification

[STIPULATION] A threefold identification acts on the torus. The quotient T^4/Z_3 is no longer a torus but an orbifold: it has points that remain fixed under the group action.

[B] The practical consequence is that every field quantity decomposes into three sectors that differ by the same phase under the identification. On a three-element fibre such an operator becomes cyclic, and a cyclic operator is diagonalised by the discrete Fourier transform. This is the structural reason why the mass sector later manages without a characteristic polynomial (A110).

[B] On admissibility: the crystallographic restriction $2\cos(2\pi/n) \in \mathbb{Z}$ permits $n \in \{1, 2, 3, 4, 6\}$ and excludes $n = 5$. Threefold symmetry is lattice-compatible, fivefold symmetry is not. This does not justify three – it constrains it.

.7 Verification Scripts

- Integrality and coordinate-independence of the winding number, shown on a closed path on T^2 under reparametrisation and continuous deformation: `python/A_Serie_Skripte/a030_windungszahl.py`
- Discreteness of the spectrum on the compact torus versus the continuous spectrum in the non-compact case, including smallest permitted step: `python/A_Serie_Skripte/a030_spektrum_diskret.py`

.8 Open Questions

First, the geometry does not say which circle is unrolled. The selection is deferred to the measurement process (A090). That is a deferral, not a result – as long as what determines the measurement process is not specified, the question is deferred and not answered.

Second, the number of fixed points of the orbifold is not evaluated here. The Z_3 action generates singular points; what physical role they play is not treated in this series.

Third, the assignment eigenvalue-mass remains a claim. That a compact space has a discrete spectrum is mathematics. That *this* spectrum is the particle masses is the physical claim of the theory and is tested against numbers only in A100 and A110.

.9 Supersedes

This document subsumes: Dok. 181 (torus justification), Dok. 188 (geometry fundamentals), Dok. 189 (torus derivation), Dok. 159 (harmonic structure), Dok. 171 (string/torus), Dok. 204 (fractal boundary condition), Dok. 302 (unit cells and partition invariance). Incorporated: P34, R56 (crystallographic restriction constrains $n = 3$, does not justify it).

.10 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.